TONE SYSTEM AND PITCH ACCENT IN STANDARD UKHRUL TANGKHUL

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ABSTRACT:

This paper investigates the phonetic and phonological properties of the tone system of Standard Ukhrul Tangkhul. While early missionary grammars treated Tangkhul as non-tonal or merely noted pitch variation, instrumental F0 analysis of monosyllabic minimal triplets reveals three distinct contrastive lexical tones: Tone 1 (High Level, 55), Tone 2 (Mid Level, 33), and Tone 3 (Low Falling, 21). We examine tone perturbation caused by syllable-final consonants and document tone sandhi processes occurring in nominal compounding and verbal suffixation.

Keywords: Tone, Pitch Accent, Tone Sandhi, Ukhrul, Fundamental Frequency, Khayi, NEILS

1. LEXICAL TONE CONTRASTS: MINIMAL TRIPLETS

The existence of tone in Tangkhul is conclusively demonstrated by numerous minimal triplets where segmental phonemes are identical but pitch contour differentiates semantic meaning. For instance, the monosyllabic frame /thang/ carries three distinct lexical meanings depending on whether it is uttered with High (55), Mid (33), or Low (21) pitch.

Tangkhul Linguistic Data & Analysis:

* **thang (Tone 1: High Level 55) : day, daylight period**
  Analysis: F0 fundamental frequency ~ 210 Hz.
* **thang (Tone 2: Mid Level 33) : to lift up, raise high**
  Analysis: F0 fundamental frequency ~ 165 Hz.
* **thang (Tone 3: Low Falling 21) : knife, machete, dao**
  Analysis: F0 fundamental frequency ~ 130 Hz.

2. TONAL ACOUSTIC MEASUREMENTS & F0 CONTOURS

Pitch contours were tracked across 10 native speakers (5 male, 5 female). Tone 1 exhibits a sustained flat high contour occupying the upper 20% of the speaker's vocal range. Tone 2 runs smoothly along the middle register. Tone 3 exhibits a slight initial rise followed by a steep descent, particularly before pause or sentence boundaries. Sonorant codas carry the pitch contour smoothly, whereas stop codas trigger abrupt glottalization that truncates the contour.

Tangkhul Linguistic Data & Analysis:

* **ka (High) vs ka (Mid) vs ka (Low) : to climb vs to bite vs to dry up**
  Analysis: Monosyllabic vowel minimal set.
* **sØng (High) vs seng (Mid) vs sŁng (Low) : clean vs liver vs iron tool**
  Analysis: Sonorant coda minimal set.

3. TONE SANDHI IN COMPOUNDING

When two monosyllabic words combine into a compound noun, the tone of the second element frequently undergoes sandhi. A Low tone following a High tone is raised to Mid to prevent extreme pitch leaps within a single phonological word. These tone sandhi rules operate systematically across both spoken discourse and traditional musical poetry.

Tangkhul Linguistic Data & Analysis:

*** shim (Mid) + khor (Low) -> shimkhor [33-33] : courtyard**

Analysis: Low tone raising following Mid stem.

*** lui (High) + ram (Low) -> luiram [55-33] : cultivated field territory**

Analysis: Low tone raising following High stem.

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